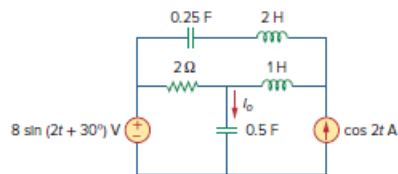


Problem

Using nodal analysis, find $i_o(t)$ in the circuit in Fig. 10.60.

Figure 10.60:



Step-by-step solution

Step 1 of 6

Refer to Figure 10.60 in the textbook.

Consider $\omega = 2 \text{ rad/s}$.

Determine the polar form of $8 \sin(2t + 30^\circ)$.

Consider $\cos(2t) = 1 \angle 0^\circ$.

$$\begin{aligned} 8 \sin(2t + 30^\circ) &= 8 \cos(2t + 30^\circ - 90^\circ) \\ &= 8 \angle -60^\circ \end{aligned}$$

Determine the reactance of the inductor for $L = 1 \text{ H}$.

$$\begin{aligned} X_L &= j\omega L \\ &= j(2)(1) \\ &= j2 \Omega \end{aligned}$$

Determine the reactance of the inductor for $L = 2 \text{ H}$.

$$\begin{aligned} X_L &= j\omega L \\ &= j(2)(2) \\ &= j4 \Omega \end{aligned}$$

Step 2 of 6

Determine the reactance of the capacitor for $C = 0.25 \text{ F}$.

$$\begin{aligned} X_C &= \left(\frac{1}{j\omega C} \right) \\ &= \left(\frac{1}{j(2)(0.25)} \right) \\ &= -j2 \Omega \end{aligned}$$

Determine the reactance of the capacitor for $C = 0.5 \text{ F}$.

$$\begin{aligned} X_C &= \left(\frac{1}{j\omega C} \right) \\ &= \left(\frac{1}{j(2)(0.5)} \right) \\ &= -j \Omega \end{aligned}$$

Step 3 of 6

Redraw the circuit diagram in frequency domain.

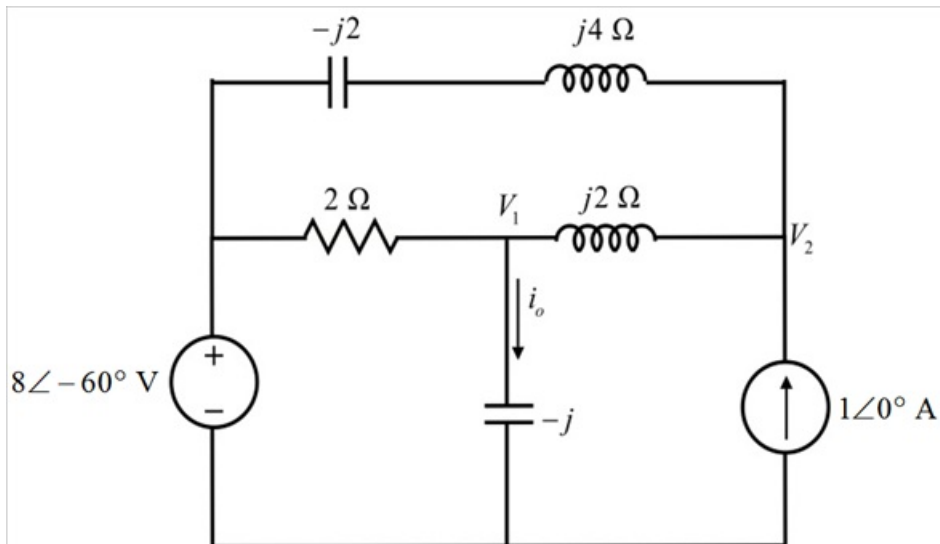


Figure 1

Step 4 of 6

Write the Kirchhoff's current law equation at node V_1 .

$$\frac{V_1 - 8\angle -60^\circ}{2} + \frac{V_1}{-j} + \frac{V_1 - V_2}{j2} = 0$$

$$V_1 \left(\frac{1}{2} + \frac{1}{-j} + \frac{1}{j2} \right) + V_2 \left(\frac{-1}{j2} \right) = \frac{8\angle -60^\circ}{2}$$

$$V_1 (0.5 + j0.5) + V_2 (j0.5) = 4\angle -60^\circ$$

$$V_1 (0.5 + j0.5) + V_2 (j0.5) = 2 - j3.464 \dots\dots (1)$$

Write the Kirchhoff's current law at node V_2 .

$$\frac{V_2 - V_1}{j2} + \frac{V_2 - 8\angle -60^\circ}{-j2 + j4} - 1\angle 0^\circ = 0$$

$$\left(\frac{-1}{j2} \right) V_1 + \left(\frac{1}{j2} + \frac{1}{j2} \right) V_2 - \frac{8\angle -60^\circ}{j2} - 1 = 0$$

$$j0.5V_1 - jV_2 - (-3.464 - j2) - 1 = 0$$

$$j0.5V_1 - jV_2 = -3.464 - j2 + 1$$

$$j0.5V_1 - jV_2 = -2.464 - j2$$

$$0.5V_1 - V_2 = \frac{-2.464 - j2}{j}$$

$$0.5V_1 - V_2 = -2 + j2.464$$

$$V_2 = 0.5V_1 + 2 - j2.464 \dots\dots (2)$$

Step 5 of 6

Substitute $0.5V_1 + 2 - j2.464$ for V_2 in equation (1).

$$V_1(0.5 + j0.5) + (0.5V_1 + 2 - j2.464)(j0.5) = 2 - j3.464$$

$$V_1(0.5 + j0.5 + j0.25) + 1.232 + j = 2 - j3.464$$

$$V_1(0.5 + j0.5 + j0.25) = 2 - j3.464 - 1.232 - j$$

$$V_1(0.5 + j0.75) = 0.768 - j4.464$$

$$V_1 = \frac{0.768 - j4.464}{0.5 + j0.75}$$

$$V_1 = 5.025 \angle -136.55^\circ \text{ V}$$

Step 6 of 6

Calculate the value of I_o .

$$I_o = \frac{V_1}{-j1}$$

Substitute $5.025 \angle -136.55^\circ \text{ V}$ for V_1 .

$$\begin{aligned} I_o &= \frac{5.025 \angle -136.55^\circ}{-j1} \\ &= 5.025 \angle -46.55^\circ \text{ A} \end{aligned}$$

Calculate the current in time domain.

$$i_o(t) = 5.025 \cos(2t - 46.55^\circ) \text{ A}$$

Therefore, the current $i_o(t)$ is $\boxed{5.025 \cos(2t - 46.55^\circ) \text{ A}}$.